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(54) **TRITYL-BASED ALCOHOL PROTECTING GROUP**

(71) Applicant: **SynCrest Inc.**, Kanagawa (JP)

(72) Inventors: **Pengyu XU**, Kanagawa (JP); **Shin YONEYAMA**, Kanagawa (JP)

(73) Assignee: **SynCrest Inc.**, Kanagawa (JP)

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(57) **ABSTRACT**

An object of the present invention is to provide a novel trityl-based alcohol protecting group that is useful in synthesizing nucleic acids. Provided is a trityl-based alcohol protecting group.

TRITYL-BASED ALCOHOL PROTECTING GROUP

TECHNICAL FIELD

[0001] The present invention relates to a trityl-based alcohol protecting group.

BACKGROUND ART

[0002] NPL 1 and NPL 2 disclose a method for synthesizing a nucleic acid using a 4,4'-dimethoxytrityl group (a DMTr protecting group).

CITATION LIST

Non-Patent Literature

[0003] NPL 1: Tetrahedron Letters, 22, 1859-1862, 1981

[0004] NPL 2: J. Am. Chem. Soc., 85, 3821-3827, 1963

SUMMARY OF INVENTION

Technical Problem

[0005] An object of the present invention is to provide a novel trityl-based alcohol protecting group that is useful in synthesizing nucleic acids.

Solution to Problem

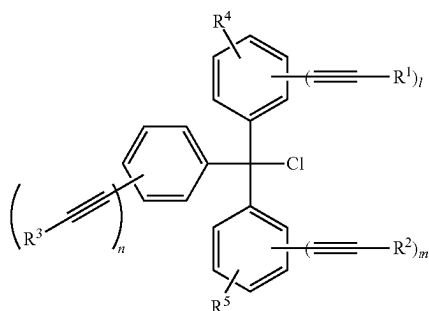
[0006] The present inventors have conducted extensive research to achieve the above object. As a result, the present inventors found that by introducing at least one triple bond into a protecting group, such as a 4,4'-dimethoxytrityl group (a DMTr protecting group), a novel trityl-based alcohol protecting group that is useful in synthesizing nucleic acids can be produced.

[0007] The present inventors conducted further research based on this finding and have accomplished the present invention.

[0008] Specifically, the present invention includes the following trityl-based alcohol protecting groups.

Item 1

[0009] A compound represented by the following formula (1):

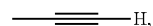


(1)

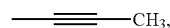
(wherein

[0010] R^1 , R^2 , and R^3 are same or different, and each represents a hydrogen atom, an alkyl group, a trimeth-

ylsilyl group (TMS, $-\text{Si}(\text{CH}_3)_3$), an acetyl group (Ac), a benzoyl group (Bz), an ester group, an alkyl ester group, an aryl group, an alkyl alcohol group, a nitrile group (CN), a group represented by



or a group represented by



[0011] wherein l , m , and n each represent an integer of 0, 1, or 2 (provided that l , m , and n are not 0 at the same time), and

[0012] R^4 and R^5 are the same or different, and each represents an electron-donating group).

[0013] As a trityl-based alcohol protecting group, the compound represented by formula (1) of the present invention is excellent in controlling the ease of deprotection. The compound represented by formula (1) of the present invention enables improvement of a substituent, such as a DMTr protecting group, and adjustment of the deprotection conditions to thereby increase the yield of the nucleic acid synthesized. Based on the triple bond introduced into the trityl-based alcohol protecting group, the compound represented by formula (1) of the present invention can change cationic properties of the trityl moiety and allows the protecting group to be readily removed for deprotection.

[0014] The triple bond introduced into the compound represented by formula (1) of the present invention in the form of a trityl-based alcohol protecting group can be detected by Raman spectroscopy. Based on the trityl-based alcohol protecting group, the compound represented by formula (1) of the present invention allows for detection of quantifiable peaks in the silent region and enables quantification of the protecting group without being disturbed by other impurities and is excellent in tracking various reactions.

Advantageous Effects of Invention

[0015] The present invention can newly provide a trityl-based alcohol protecting group that is useful in synthesizing nucleic acids.

DESCRIPTION OF EMBODIMENT

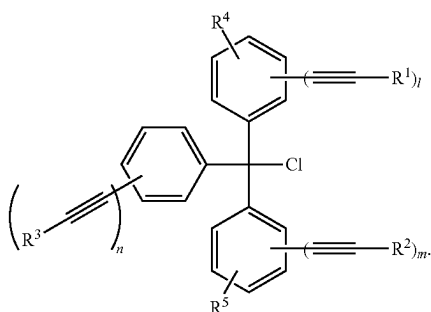
[0016] The present invention is described in detail below.

[0017] In the present specification, the terms “comprise” and “contain” encompass the concepts of comprising, consisting essentially of, and consisting of.

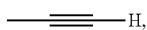
[0018] In the present specification, the numerical range indicated by “A to B” means A or more and B or less, unless otherwise specified.

[1] Trityl-Based Alcohol Protecting Group

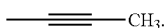
[0019] The trityl-based alcohol protecting group of the present invention is a compound represented by the following formula (1):



[0020] In formula (1), R^1 , R^2 , and R^3 are same or different, and each represents a hydrogen atom, an alkyl group, a trimethylsilyl group, an acetyl group, a benzoyl group, an ester group, an alkyl ester group, an aryl group, an alkyl alcohol group, a nitrile group, a group represented by



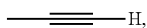
or a group represented by formula (6)



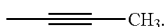
[0021] In formula (1), l , m , and n each represent an integer of 0, 1, or 2, provided that l , m , and n are not 0 at the same time.

[0022] In formula (1), R^4 and R^5 may be same or different, and each represents an electron-donating group. R^1 , R^2 , and R^3

[0023] In formula (1), R^1 , R^2 , and R^3 are same or different, and each represents a hydrogen atom (H), an alkyl group, a trimethylsilyl group (TMS, $\text{---Si(CH}_3)_3$), an acetyl group (Ac), a benzoyl group (Bz), an ester group, an alkyl ester group, an aryl group, an alkyl alcohol group, a nitrile group (CN), a group represented by



or a group represented by



[0024] The alkyl group is preferably, for example, a methyl group (---CH_3 , Me), an ethyl group (Et), an n-butyl group (n-Bu), or a t-butyl group (t-Bu).

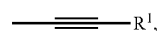
[0025] The ester group is preferably, for example, a methyl ester group (COOMe) or an ethyl ester group (COOEt).

[0026] The alkyl ester group is preferably, for example, a 3-carboxypropyl group ($((\text{CH}_2)_3\text{COOH})$) or a 3-methyloxycarbonylpropyl group ($((\text{CH}_2)_3\text{COOMe})$).

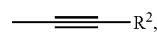
[0027] The aryl group is preferably, for example, a phenyl group (Ph), a 4-methyloxycarbonylbenzene group

(4-COOMe- C_6H_4), a 4-formylbenzene group (4-CHO- C_6H_4), a 4-methylhydroxybenzene group (4- CH_2OH - C_6H_4), or a 3,5-dimethoxybenzene group (3,5-OMe- C_6H_4). [0028] The alkyl alcohol group (e.g., CH_2OH , $(\text{CH}_2)_2\text{OH}$, $(\text{CH}_2)_3\text{OH}$), $\text{CH}_2\text{OCO}_2(4\text{-NO}_2\text{-C}_6\text{H}_4)$, or $(\text{CH}_2)_3\text{OCO}_2(4\text{-NO}_2\text{-C}_6\text{H}_4)$ is preferably, for example, a methyl hydroxyl group (CH_2OH), an ethyl hydroxyl group ($(\text{CH}_2)_2\text{OH}$), a propyl hydroxyl group ($(\text{CH}_2)_3\text{OH}$), a methyl(4-nitrobenzo)oxycarbonyl group ($\text{CH}_2\text{OCO}_2(4\text{-NO}_2\text{-C}_6\text{H}_4)$), or a propyl (4-nitrobenzo)oxycarbonyl group ($(\text{CH}_2)_3\text{OCO}_2(4\text{-NO}_2\text{-C}_6\text{H}_4)$).

[0029] In formula (1), the substituent of a triple bond (alkyne) including R^1 :

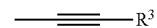


[0030] the substituent of a triple bond (alkyne) including R^2 :



and

[0031] the substituent of a triple bond (alkyne) including R^3 :



may be bonded at any of the ortho-, meta-, or para-positions.

[0032] In formula (1), the substituent of a triple bond including R^1 , the substituent of a triple bond including R^2 , and the substituent of a triple bond including R^3 are each bonded, for example, at a para-position with respect to the carbon atom of the benzene ring bonded to the central carbon atom (the “C” in “ $\text{---C(C}_6\text{H}_5)_3$ ”) of the trityl group ($\text{---C(C}_6\text{H}_5)_3$); and

[0033] l , m , and n each represent an integer of 0, 1, or 2,

[0034] provided that l , m , and n are not 0 at the same time,

[0035] l is the number of substituents in the triple bond including R^1 (alkyne),

[0036] m is the number of substituents in the triple bond including R^2 (alkyne), and

[0037] n is the number of substituents in the triple bond including R^3 (alkyne).

[0038] In formula (1), at least one of R^1 , R^2 , and R^3 is preferably a trimethylsilyl group. R^4 and R^5

[0039] In formula (1), R^4 and R^5 may be same or different, and each represents an electron-donating group.

[0040] The electron-donating group is preferably

[0041] a linear or branched alkyl group, such as methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, or tert-butyl;

[0042] a cycloalkyl group, such as cyclopropyl, cyclobutyl, or cyclopentyl;

[0043] an alkenyl group, such as vinyl, allyl, or 1-propenyl;

[0044] a cycloalkenyl group, such as 1-cyclopentenyl or 1-cyclohexenyl;

[0045] an aryl group, such as phenyl;

[0046] an aralkyl group, such as benzyl or phenethyl;

[0047] an alkoxy group, such as methoxy, ethoxy, n-propoxy, iso-propoxy, n-butoxy, or tert-butoxy;

[0048] an aryloxy group, such as phenyloxy;

[0049] a hydroxy group;

[0050] an amino group;

[0051] an amino group having one or two alkyl groups, such as methylamino, ethylamino, n-propylamino, dimethylamino, or diethylamino;

[0052] an amino group having one or two aryl groups, such as phenylamino; or a hydroxyamino group.

[0053] The electron-donating group is preferably an alkyl sulfide group, such as a methyl sulfide group (methylthio group), an ethyl sulfide group, an n-propyl sulfide group, an isopropyl sulfide group, an n-butyl sulfide group, an isobutyl sulfide group, or a tert-butyl sulfide group.

[0054] The electron-donating group is more preferably, for example, a linear or branched alkyl group, an alkoxy group, a hydroxy group, an amino group, or an amino group having one or two alkyl group, even more preferably an alkoxy group, a hydroxy group, or an amino group, and particularly preferably an alkoxy group having 1 to 4 carbon atoms.

[0055] The electron-donating group is preferably an alkoxy group, such as methoxy ($-\text{OMe}$), ethoxy ($-\text{OEt}$), n-propoxy ($-\text{OnPr}$), isopropoxy ($-\text{OiPr}$), n-butoxy ($-\text{OnBu}$), or tert-butoxy ($-\text{OtBu}$).

[0056] The electron-donating group is particularly preferably a linear or branched alkyl group, such as methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, or tert-butyl.

[0057] The electron-donating group is particularly preferably an alkylsulfide group, such as a methylsulfide group (e.g., a methylthio group, $-\text{S-Me}$, $-\text{S-CH}_3$) or an ethylsulfide group ($-\text{S-Et}$).

[0058] In formula (1), R^4 and R^5 may be each bonded at any of the ortho-, meta-, or para-positions.

[0059] In formula (1), R^4 and R^5 are each preferably, for example, bonded at a para-position with respect to the carbon atom of a benzene ring bonded to the central carbon atom (" $-\text{C}$ " in " $-\text{C}(\text{C}_6\text{H}_5)_3$ ") of the trityl group ($-\text{C}(\text{C}_6\text{H}_5)_3$).

[0060] The trityl-based alcohol protecting group of the present invention is a protecting group, such as a 4,4'-dimethoxytrityl group (a DMTr protecting group), that has at least one triple bond introduced into the protecting group. The triple bond introduced into the trityl-based alcohol protecting group conjugates with a DMTr protecting group or like protecting group to thereby change cationic properties of the trityl moiety of the DMTr protecting group. Due to this feature, the trityl-based alcohol protecting group of the present invention can control the ease of removal of the protecting group from a target nucleic acid.

[0061] The DMTr protecting group or like protecting group of the trityl-based alcohol protecting group changes in stability of the cation. For example, under weak acidic

conditions, the protecting group forms a trityl-based alcohol protecting group that can be easily or instantaneously removed.

[0062] The triple bonds introduced into the trityl-based alcohol protecting group of the present invention produce quantifiable peaks in the silent region in Raman spectroscopy. Due to this feature, the trityl-based alcohol protecting group of the present invention allows for quantitative determination of the protecting group without being interfered with by other impurities.

[0063] The trityl-based alcohol protecting group of the present invention is useful in producing nucleic acids.

[0064] Embodiments of the present invention are described above; however, the present invention is not limited to these examples. Needless to say, the present invention can be implemented in various forms within the scope not departing from the gist of the present invention.

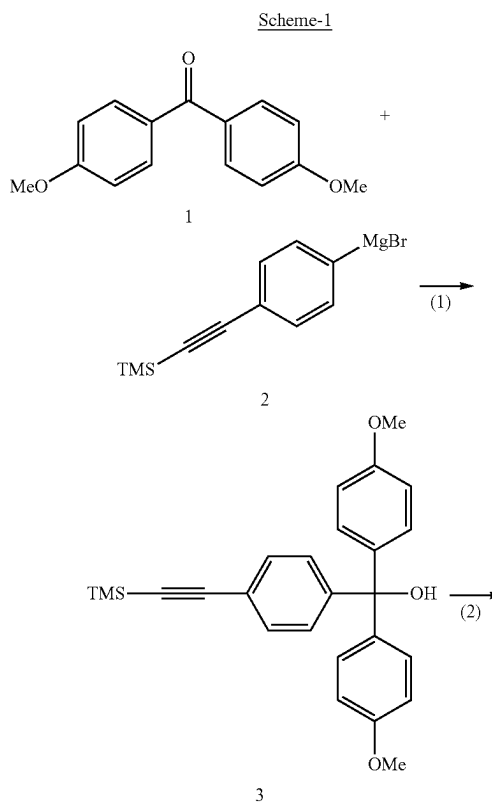
EXAMPLES

[0065] Embodiments of the present invention are described in more detail below based on Examples.

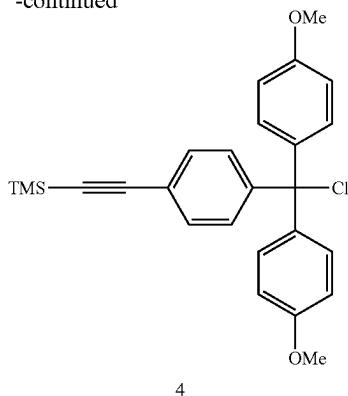
[0066] The present invention is not limited to these Examples.

EXAMPLES

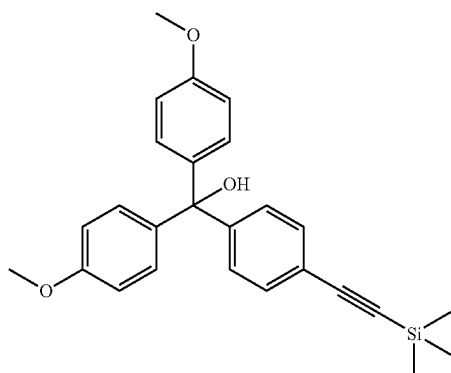
[1] Production 1 of Trityl-Based Alcohol Protecting Group (Scheme 1)



-continued



(1-1) Production of Compound 3 in Scheme 1



[0067] Under an argon atmosphere, magnesium powder (720 mg, 29.6 mmol) and iodine (2 items) were placed in a 100 mL round-bottom flask, and 23 mL of tetrahydrofuran (THF) was added. The resulting mixture was stirred at room temperature. A solution (8.0 mL) of 4-(bromophenylethynyl)trimethylsilane (a precursor of Compound 2 in Scheme 1) (5.00 g, 19.7 mmol) in THF was added dropwise thereto and the resulting mixture was stirred at room temperature for 2 hours to prepare a Grignard reagent of 4-(bromophenylethynyl)trimethylsilane (Compound 2 in Scheme 1).

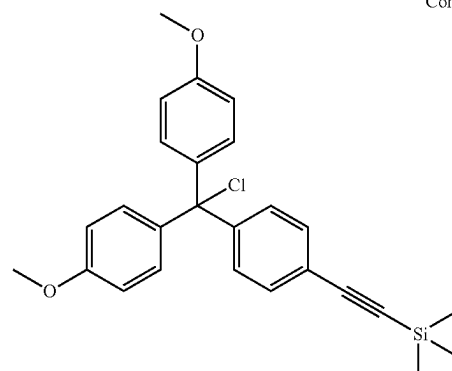
[0068] 4,4'-Dimethoxybenzophenone (Compound 1 in Scheme 1) (4.29 g, 17.7 mmol) was added to this reaction liquid, and the resulting mixture was stirred at room temperature for 3 days. After completion of the reaction was confirmed, an aqueous ammonium chloride solution was added to the reaction liquid, and the resulting mixture was extracted with ethyl acetate. The organic layer was dried over magnesium sulfide and concentrated.

[0069] The residue was purified by silica gel column chromatography (hexane/ethyl acetate=19/1→9/1) to obtain the target compound (Compound 3 in Scheme 1) (4.63 g, 11.1 mmol) as a white solid.

[0070] $^1\text{H-NMR}$ (CDCl_3): δ 7.42-7.38 (m, 2H), 7.25-7.21 (m, 2H), 7.16-7.11 (m, 4H), 6.85-6.80 (m, 4H), 3.80 (s, 6H), 2.66 (s, 1H), 0.24 (s, 9H)

[0071] $^{13}\text{C-NMR}$ (CDCl_3): δ 158.8, 147.6, 139.0, 131.5, 129.1, 127.6, 121.8, 113.3, 104.9, 94.3, 81.3, 55.3, 0.0

(1-2) Production of Compound 4 (Trityl-Based Alcohol Protecting Group) in Scheme 1



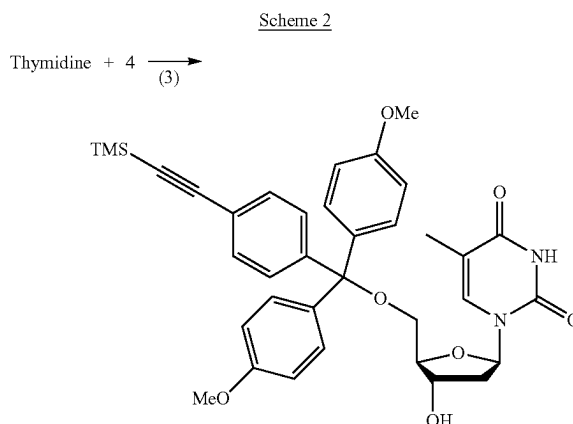
[0072] Thionyl chloride (0.17 mL, 2.4 mmol) and N, N-dimethylformamide (DMF) (1 drop) were added to 4-trimethylsilylethynyl-4',4''-dimethoxytrityl alcohol (Compound 3 in Scheme 1) (0.20 g, 0.48 mmol), and the resulting mixture was stirred at 45° C. for 1.5 hours.

[0073] The stirred mixture was then concentrated and the residue of the target compound was obtained as a red solid (Compound 4 (trityl-based alcohol protecting group) in Scheme 1).

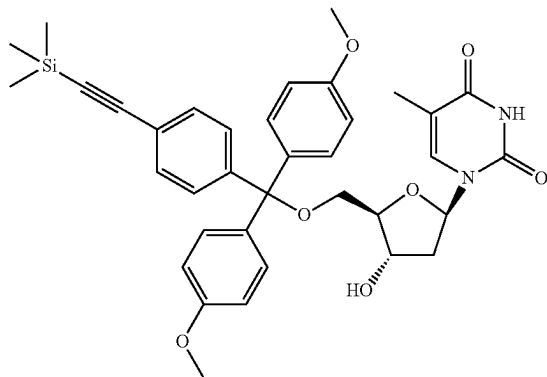
[0074] $^1\text{H-NMR}$ (CDCl_3): δ 7.41-7.37 (m, 2H), 7.22-7.18 (m, 2H), 7.15-7.10 (m, 4H), 6.84-6.79 (m, 4H), 3.82 (s, 6H), 0.25 (s, 9H)

[0075] $^{13}\text{C-NMR}$ (CDCl_3): δ 159.3, 146.1, 137.4, 131.4, 131.1, 129.7, 122.7, 113.2, 113.1, 104.7, 95.3, 82.4, 55.5, 0.1

(1-3) Protection of Nucleic Acid by Trityl-Based Alcohol Protecting Group (Scheme 2)



Nucleic Acid Protected with a Trityl-Based Alcohol-Protecting Group



[0076] Pyridine (2.0 mL) was added to thymidine (nucleic acid) (90 mg, 0.37 mmol) and 4-trimethylsilylethynyl-4',4''-dimethoxytrityl chloride (Compound 4 and a trityl-based alcohol protecting group in Schemes 1 and 2) (178 mg, 0.41 mmol), and the resulting mixture was stirred at 50° C. overnight. The stirred mixture was then concentrated, and an aqueous ammonium chloride solution was added. The resulting mixture was extracted with ethyl acetate, and the organic layer was dried over magnesium sulfide and concentrated.

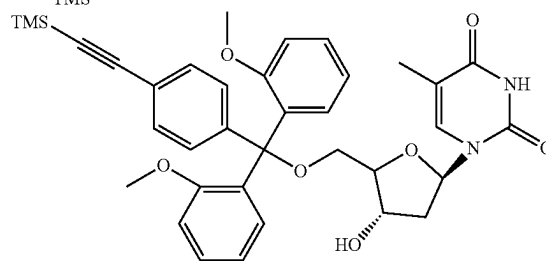
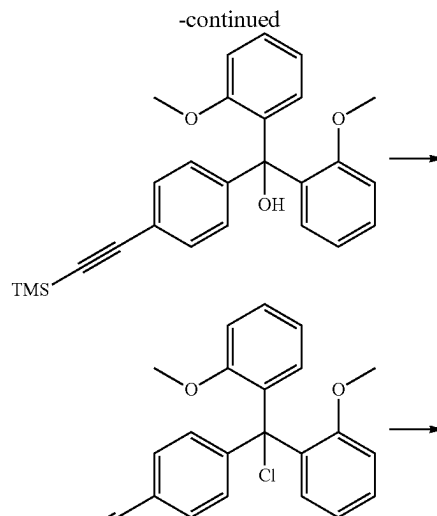
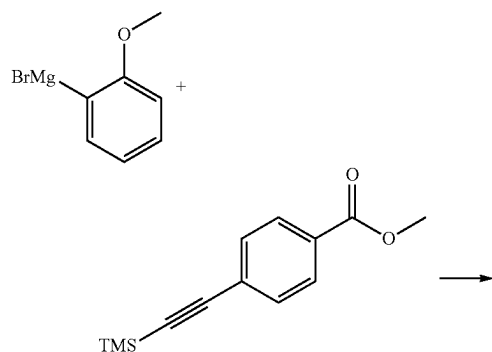
[0077] The residue was purified by silica gel column chromatography (hexane/ethyl acetate=1/1→0/1) to obtain the target compound (90 mg, 0.14 mmol) as a white solid (a nucleic acid protected with a trityl-based alcohol protecting group).

[0078] ¹H-NMR (DMSO-d₆): δ 11.33 (s, 1H), 7.45 (d, 1H, J=1.2 Hz), 7.39-7.37 (m, 4H), 7.25-7.22 (m, 4H), 6.91-6.88 (m, 4H), 6.20 (t, 1H, J=6.6 Hz), 5.30 (d, 1H, J=4.5 Hz), 4.28-4.22 (m, 1H), 3.89-3.85 (m, 1H), 3.74 (s, 6H), 3.23-3.12 (m, 2H), 2.27-2.13 (m, 2H), 1.50 (d, 3H, J=1.2 Hz), 0.21 (s, 9H)

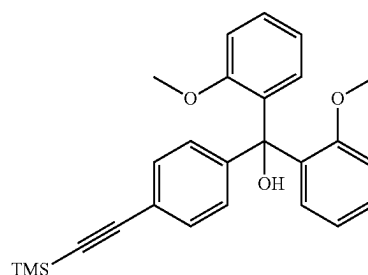
[0079] ¹³C-NMR (DMSO-d₆): δ 163.7, 158.3, 150.4, 145.8, 135.7, 134.9, 134.5, 131.2, 129.8, 129.7, 127.9, 120.6, 113.3, 109.6, 104.9, 94.3, 85.6, 85.3, 83.8, 70.5, 64.0, 55.1, 39.3, 11.8, -0.1

[2] Production 2 of Trityl-Based Alcohol Protecting Group (Scheme 3)

Scheme 3



(2-1) Production of 4-trimethylsilylethynyl-2',2''-dimethoxytrityl Alcohol



[0080] Under argon atmosphere, magnesium powder (306 mg, 12.6 mmol) and iodine (1 item) were placed in a 100 mL eggplant flask, and THF (10 mL) was added. The resulting mixture was stirred at room temperature. A solution (3.5 mL) of 2-bromoanisole (1.57 g, 8.40 mmol) in THF was added dropwise thereto, and the resulting mixture was stirred at a temperature of 40° C. to 50° C. for 1 hour.

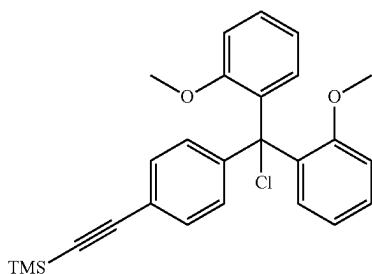
[0081] Methyl 4-(trimethylsilylethynyl)benzoate (888 mg, 3.82 mmol) was added to this reaction liquid, and the resulting mixture was stirred at room temperature for 1 hour. After completion of the reaction was confirmed, 1N hydrochloric acid was added to the reaction liquid, and the resulting mixture was extracted with ethyl acetate. The organic layer was dried over magnesium sulfide and concentrated.

[0082] The residue was purified by silica gel column chromatography (hexane/ethyl acetate=19/1→4/1) to obtain the target compound (1.45 g, 3.48 mmol) as a yellow solid.

[0083] $^1\text{H-NMR}$ (CDCl_3): δ 7.38-7.34 (m, 2H), 7.32-7.26 (m, 2H), 7.22-7.18 (m, 2H), 7.00-6.97 (m, 2H), 6.93-6.85 (m, 4H), 5.26 (s, 1H), 3.52 (s, 6H), 0.24 (s, 9H)

[0084] $^{13}\text{C-NMR}$ (CDCl_3): δ 157.5, 147.7, 133.6, 131.1, 129.5, 129.0, 127.8, 121.1, 120.6, 112.4, 105.7, 93.7, 80.9, 55.6, 0.2

(2-2) Production of 4-trimethylsilylethynyl-2',2"-dimethoxytrityl Chloride

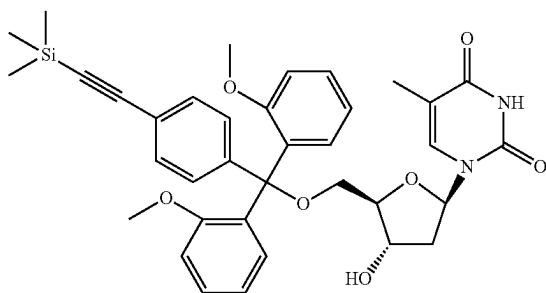


[0085] Ethyl acetate (2.4 mL) and thionyl chloride (0.35 mL, 4.8 mmol) were added to 4-trimethylsilylethynyl-2',2"-dimethoxytrityl alcohol (0.20 g, 0.48 mmol), and the resulting mixture was stirred at 45° C. for 4 hours. The stirred mixture was concentrated to obtain a residue of the target compound as a beige solid.

[0086] $^1\text{H-NMR}$ (CDCl_3): δ 7.37-7.27 (m, 6H), 7.15-7.11 (m, 2H), 6.92-6.84 (m, 4H), 3.51 (s, 6H), 0.25 (s, 9H)
 $^{13}\text{C-NMR}$ (CDCl_3): δ 157.7, 145.0, 132.0, 130.7, 130.1, 129.8, 129.2, 121.5, 120.1, 112.8, 105.3, 94.3, 80.9, 55.6, 0.1

(2-3) Protection of Nucleic Acid by Trityl-Based Alcohol Protecting Group

5'-O-(4-Trimethylsilylethynyl-2',2"-dimethoxytrityl)-thymidine



[0087] Pyridine (1.0 mL) was added to thymidine (67 mg, 0.28 mmol) and 4-trimethylsilylethynyl-2',2"-dimethoxytrityl chloride (190 mg, 0.44 mmol), and the resulting mixture was stirred at 50° C. for 3 hours and then concentrated. After adding water, the resulting mixture was extracted with ethyl acetate. The organic layer was dried over magnesium sulfide and concentrated.

[0088] The residue was purified by silica gel column chromatography (hexane/ethyl acetate 1/1→0/1) to obtain the target compound (21 mg, 0.033 mmol) as a white solid.

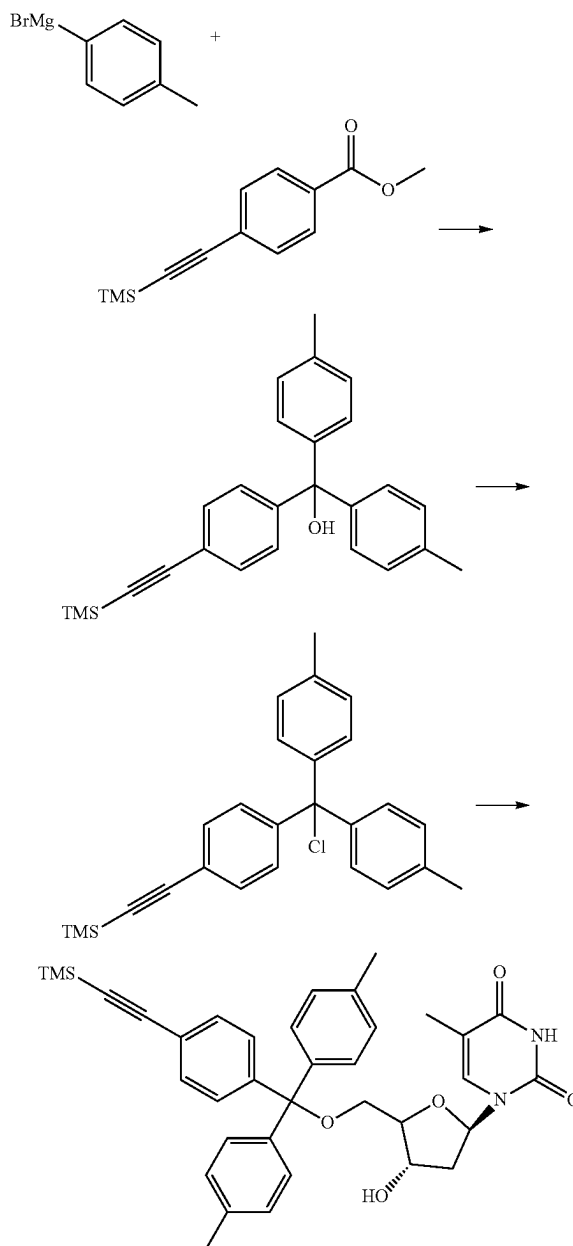
[0089] $^1\text{H-NMR}$ (DMSO-d_6): δ 11.33 (s, 1H), 7.49 (d, 1H, $J=0.9$ Hz), 7.43-7.39 (m, 1H), 7.35-7.24 (m, 7H), 7.01-6.90

(m, 4H), 6.20 (t, 1H, $J=6.6$ Hz), 5.31 (d, 1H, $J=4.5$ Hz), 4.38-4.33 (m, 1H), 3.91-3.88 (m, 1H), 3.40 (s, 3H), 3.38 (s, 3H), 3.22 (dd, 1H, $J=10.5, 4.5$ Hz), 3.13 (dd, 1H, $J=10.5, 2.7$ Hz), 2.25-2.08 (m, 2H), 1.42 (d, 3H, $J=0.9$ Hz), 0.19 (s, 9H)

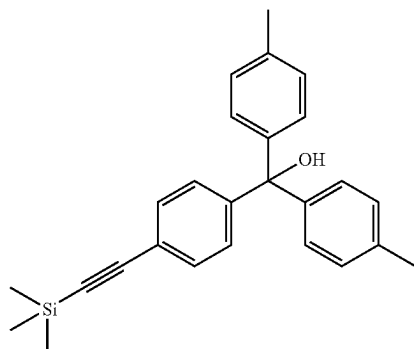
[0090] $^{13}\text{C-NMR}$ (DMSO-d_6): δ 163.7, 157.2, 156.9, 150.4, 145.8, 135.8, 130.0, 129.7, 129.3, 129.3, 129.1, 128.2, 128.0, 120.2, 120.0, 119.5, 112.3, 112.2, 109.5, 105.5, 93.4, 85.5, 84.4, 83.8, 70.5, 64.5, 54.9, 54.8, 39.6, 11.8, -0.1

[3] Production 3 of Trityl-Based Alcohol Protecting Group (Scheme 4)

Scheme 4



(3-1) Production of 4-trimethylsilylethynyl-4',4''-dimethyltrityl Alcohol



[0091] Under argon atmosphere, magnesium powder (237 mg, 9.77 mmol) and iodine (2 items) were placed in a 100 mL eggplant flask, and THF (10 mL) was added. The resulting mixture was stirred at room temperature. A solution (3.5 mL) of 4-bromotoluene (1.11 g, 6.51 mmol) in THE was added dropwise thereto, and the resulting mixture was stirred at 50° C. for 1 hour.

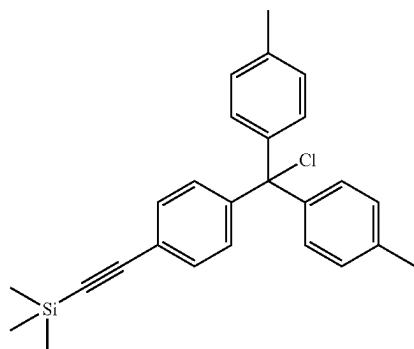
[0092] Methyl 4-(trimethylsilylethynyl)benzoate (688 mg, 2.96 mmol) was added to this reaction liquid, and the resulting mixture was stirred at room temperature for 2.5 minutes. 1N HCl was then added to the reaction liquid, and the resulting mixture was extracted with ethyl acetate. The organic layer was dried over magnesium sulfide and concentrated.

[0093] The residue was purified by silica gel column chromatography (hexane/ethyl acetate=98/2→95/5) to obtain the target compound (750 mg, 1.95 mmol) as a yellow solid.

[0094] $^1\text{H-NMR}$ (CDCl_3): δ 7.42-7.38 (m, 2H), 7.26-7.22 (m, 2H), 7.11 (s, 8H), 2.69 (s, 1H), 2.34 (s, 6H), 0.24 (s, 9H)

[0095] $^{13}\text{C-NMR}$ (CDCl_3): δ 147.6, 143.9, 137.2, 131.6, 128.8, 127.9, 127.9, 122.0, 105.1, 94.5, 81.8, 21.2, 0.1

(3-2) Production of 4-trimethylsilylethynyl-4',4''-dimethyltrityl Chloride



[0096] Thionyl chloride (0.16 mL, 2.2 mmol) was added to 4-trimethylsilylethynyl-4',4''-dimethyltrityl alcohol (170 mg, 0.44 mmol), and the resulting mixture was stirred at 45°

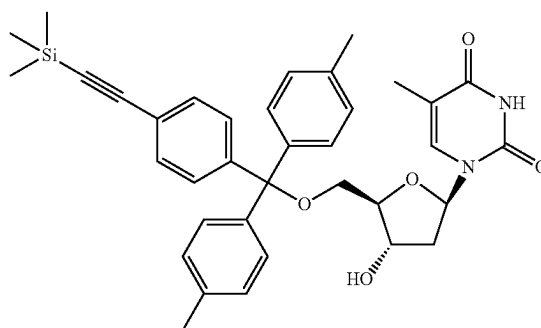
C. for 1.5 hours. The stirred mixture was then concentrated to give the target compound (177 mg, 0.44 mmol) as a white-yellow solid.

[0097] $^1\text{H-NMR}$ (CDCl_3): δ 7.40-7.36 (m, 2H), 7.21-7.17 (m, 2H), 7.09 (s, 8H), 2.36 (s, 6H), 0.25 (s, 9H)

[0098] $^{13}\text{C-NMR}$ (CDCl_3): δ 146.0, 142.3, 137.9, 131.4, 129.7, 129.7, 128.6, 122.6, 104.7, 95.2, 81.3, 21.2, 0.1

(3-3) Protection of Nucleic Acid Using a Trityl-Based Alcohol Protecting Group

5'-O-(4-Trimethylsilylethynyl-4',4''-dimethyltrityl)-thymidine



[0099] 4-Trimethylsilylethynyl-4',4''-dimethyltrityl chloride (165 mg, 0.41 mmol) and thymidine (66 mg, 0.27 mmol) were dissolved in pyridine (1.0 mL), and the resulting mixture was stirred at 50° C. overnight and then concentrated. The organic layer was extracted with ethyl acetate and dried over magnesium sulfide sulfate, and the solvent was distilled off.

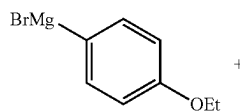
[0100] The residue was purified by column chromatography (hexane/ethyl acetate=1/1→0/1) to obtain the target compound (87.9 mg, 0.144 mmol) as a white solid.

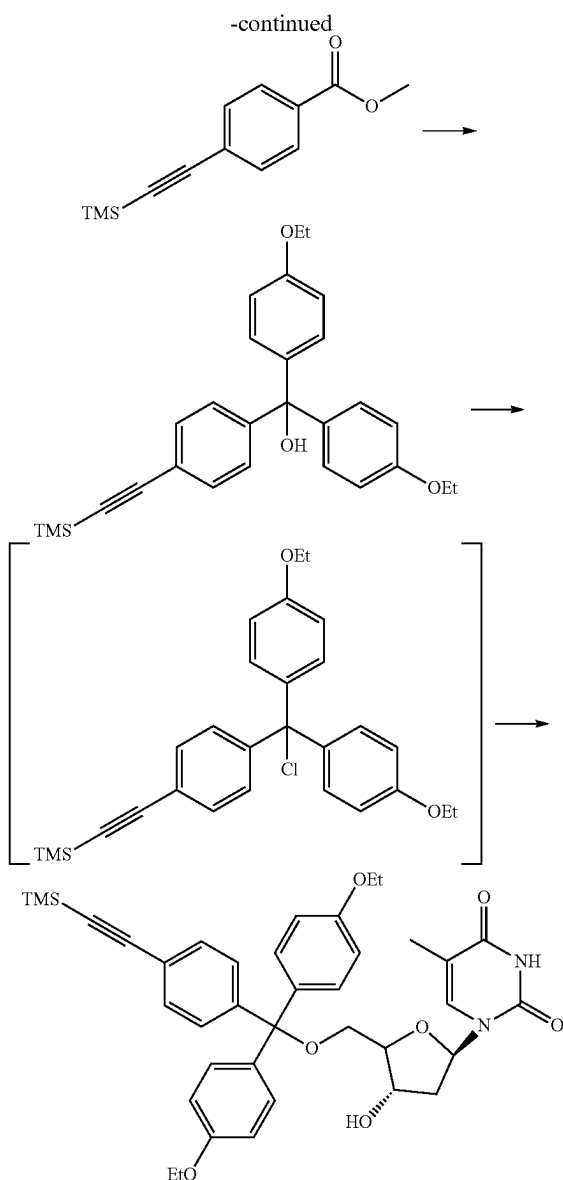
[0101] $^1\text{H-NMR}$ (DMSO-d_6): δ 11.33 (s, 1H), 7.45 (d, 1H, $J=1.2$ Hz), 7.41-7.35 (m, 4H), 7.23-7.20 (m, 4H), 7.15-7.12 (m, 4H), 6.20 (t, 1H, $J=6.6$ Hz), 5.30 (d, 1H, $J=4.5$ Hz), 4.27-4.21 (m, 1H), 3.89-3.85 (m, 1H), 3.23-3.11 (m, 1H), 2.27 (s, 6H), 2.24-2.06 (m, 2H), 1.52 (d, 3H, $J=1.2$ Hz), 0.21 (s, 9H)

[0102] $^{13}\text{C-NMR}$ (DMSO-d_6): δ 163.7, 150.4, 145.3, 140.3, 139.8, 136.5, 135.8, 131.2, 128.6, 128.4, 128.3, 128.2, 120.7, 109.6, 104.9, 94.4, 85.9, 85.3, 83.7, 70.4, 64.2, 39.5, 20.8, 11.8, -0.1

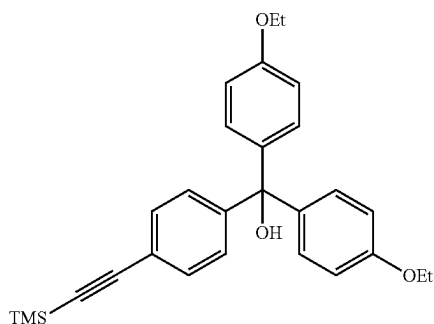
[4] Production 4 of Trityl-Based Alcohol Protecting Group (Scheme 5)

Scheme 5





(4-1) Production of 4-trimethylsilylethynyl-4',4"-dimethoxytrityl Alcohol



[0103] Under argon atmosphere, magnesium powder (237 mg, 9.77 mmol) and iodine (1 item) were placed in a 100 mL eggplant flask, and THF (10 mL) was added. The resulting mixture was stirred at room temperature. A solution (3.5 mL) of 4-bromophenol (0.928 mL, 6.51 mmol) in THF was added dropwise thereto, and the resulting mixture was stirred at 50° C. or lower for 1 hour.

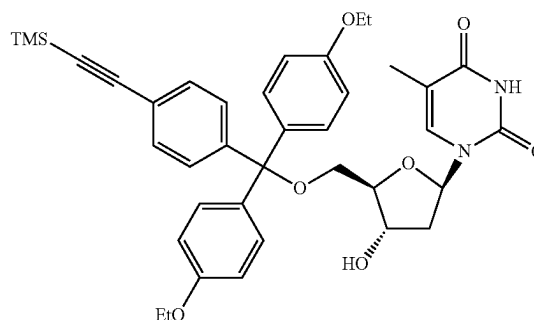
[0104] Methyl 4-(trimethylsilyl)ethynyl)benzoate (570 mg, 2.45 mmol) was added to this reaction liquid, and the resulting mixture was stirred at room temperature for 2.5 hours. 1N HCl was then added to the reaction liquid, and the resulting mixture was extracted with ethyl acetate. The organic layer was dried over magnesium sulfide and concentrated.

[1015] The residue was purified by silica gel column chromatography (hexane/ethyl acetate=98/2→95/5) to obtain the target compound (653 mg, 1.47 mmol) as a yellow oil.

[1016] ¹H-NMR (CDCl₃): δ 7.42-7.38 (m, 2H), 7.26-7.21 (m, 2H), 7.14-7.09 (m, 4H), 6.83-6.78 (m, 4H), 4.02 (q, 4H, J=6.9 Hz), 2.70 (s, 1H), 1.41 (t, 6H, J=6.9 Hz), 0.24 (s, 9H)
¹³C-NMR (CDCl₃): δ 158.3, 147.8, 139.0, 131.6, 129.2, 127.8, 121.9, 113.9, 105.1, 94.4, 81.5, 63.6, 15.0, 0.1

(4-2) Protection of Nucleic Acid Using a Trityl-Based Alcohol Protecting Group

5'-O-(4-Trimethylsilylethynyl-4',4''-diethoxyltrityl)-
thymidine



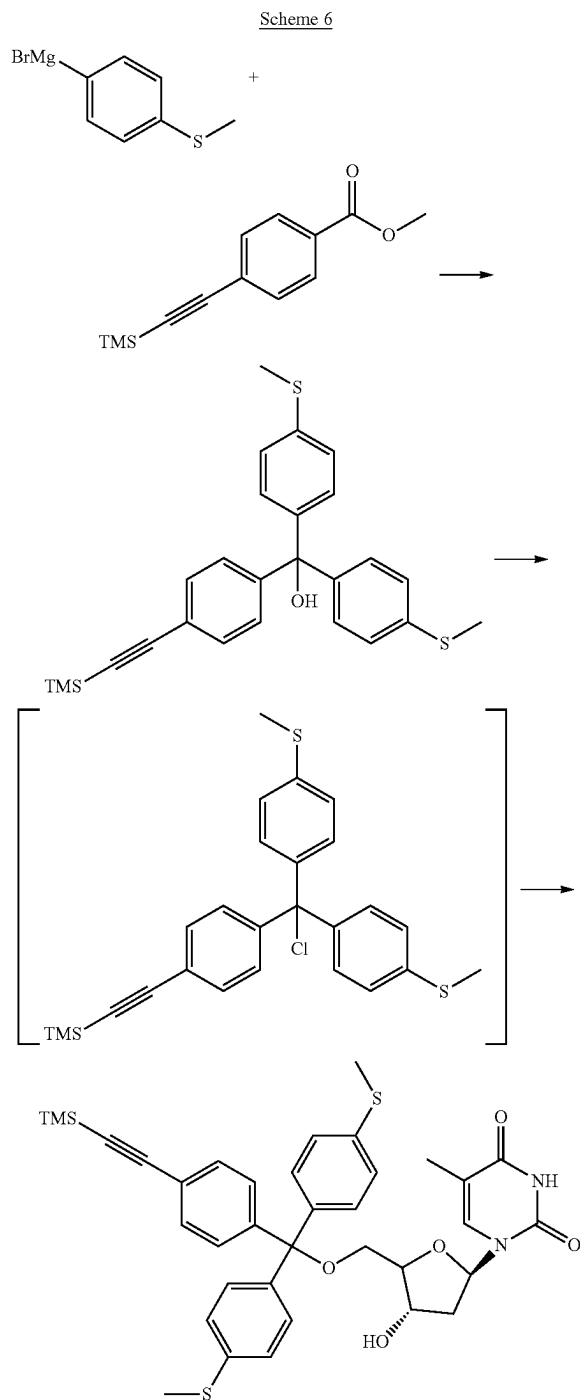
[0107] 4-Trimethylsilylethynyl-4',4''-diethoxytrityl alcohol (0.27 g, 0.60 mmol) was dissolved in ethyl acetate (3.0 mL), and thionyl chloride (0.22 mL, 3.0 mmol) was added. The resulting mixture was stirred at 45° C. for 1.5 hours. The stirred mixture was then concentrated to give a product. Subsequently, this product was dissolved in pyridine (3.0 mL), and thymidine (73 mg, 0.30 mmol) was added. The resulting mixture was stirred at 45° C. overnight.

[0108] The stirred mixture was concentrated, and the residue was purified by silica gel column chromatography (hexane/ethyl acetate=1/1→0/1) to obtain the target compound (45 mg, 0.067 mmol) as a white solid.

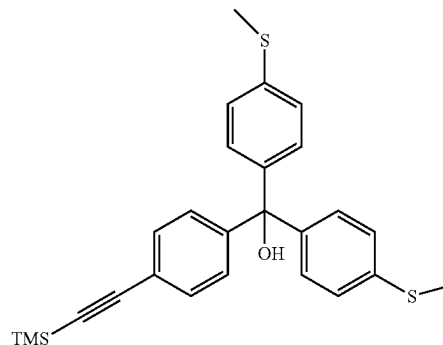
[1019] ¹H-NMR (DMSO-d₆): δ 11.32 (s, 1H), 7.45 (d, 1H, J=0.9 Hz), 7.39-7.37 (m, 4H), 7.23-7.19 (m, 4H), 6.89-6.86 (m, 4H), 6.20 (t, 1H, J=6.6 Hz), 5.30 (d, 1H, J=4.5 Hz), 4.28-4.22 (m, 1H), 4.00 (q, 4H, J=6.9 Hz), 3.89-3.85 (m, 1H), 3.22-3.12 (m, 2H), 2.27-2.08 (m, 2H), 1.50 (d, 3H, J=0.9 Hz), 1.30 (t, 6H, J=6.9 Hz), 0.21 (s, 9H)

[0110] ^{13}C -NMR ($\text{DMSO}-d_6$): δ 163.6, 157.6, 150.4, 145.8, 135.7, 134.8, 134.4, 131.2, 129.8, 129.7, 127.9, 120.6, 113.7, 109.5, 104.9, 94.3, 85.6, 85.3, 83.7, 70.4, 63.0, 39.4, 14.6, 11.8, -0.1

[5] Production 5 of Trityl-Based Alcohol Protecting Group (Scheme 6)



(5-1) Production of 4-trimethylsilylethynyl-4',4''-dimethylthiotrityl Alcohol



[0111] Under argon atmosphere, magnesium powder (237 mg, 9.77 mmol) and iodine (1 piece) were placed in a 100 mL eggplant flask, and THF (10 mL) was added. The resulting mixture was stirred at room temperature. A solution (3.5 mL) of 4-bromothiobenzene (1.32 g, 6.51 mmol) in THF was added dropwise thereto, and the resulting mixture was stirred at 50° C. or lower for 1 hour.

[0112] Methyl 4-(trimethylsilylethynyl)benzoate (690 mg, 2.96 mmol) was added to this reaction liquid, and the resulting mixture was stirred at room temperature overnight. 1N HCl was then added to the reaction liquid, the resulting mixture was extracted with ethyl acetate. The organic layer was dried over magnesium sulfide and concentrated.

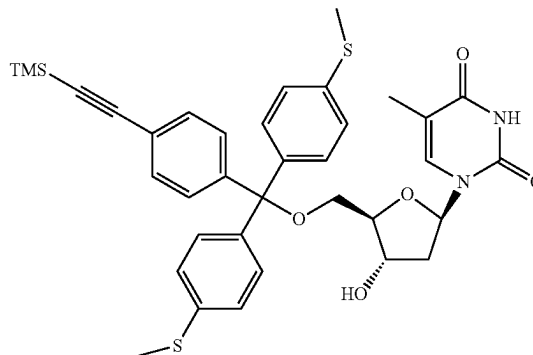
[0113] The residue was purified by silica gel column chromatography (hexane/ethyl acetate=98/2→95/5) to obtain the target compound (1.03 g, 2.29 mmol).

[0114] ^1H -NMR (CDCl_3): δ 7.43-7.38 (m, 2H), 7.24-7.12 (m, 10H), 2.69 (s, 1H), 2.47 (s, 6H), 0.24 (s, 9H)

[0115] ^{13}C -NMR (CDCl_3): δ 147.0, 143.3, 138.0, 131.7, 128.4, 127.8, 126.1, 122.3, 104.9, 94.8, 81.5, 15.8, 0.1

(5-2) Protection of Nucleic Acid Using a Trityl-Based Alcohol Protecting Group

5'-O-(4-Trimethylsilylethynyl-4',4''-dimethylthiotrityl)-thymidine



[0116] 4-Trimethylsilylethynyl-4',4''-dimethylthiotrityl alcohol (0.30 g, 0.67 mmol) was dissolved in ethyl acetate (3.2 mL), and thionyl chloride (0.24 mL, 3.4 mmol) was

added. The resulting mixture was stirred at 45° C. for 1.5 hour and then concentrated to obtain a product. Subsequently, this product was dissolved in pyridine (3.0 mL), and thymidine (73 mg, 0.30 mmol) was added. The resulting mixture was stirred at 45° C. overnight.

[0117] The stirred mixture was concentrated, and the residue was purified by silica gel column chromatography (hexane/ethyl acetate=1/1→0/1) to obtain the target compound (115 mg, 0.17 mmol) as a white solid.

[0118] ¹H-NMR (DMSO-d₆): δ 11.33 (s, 1H), 7.44 (d, 1H, J=1.2 Hz), 7.43-7.36 (m, 4H), 7.29-7.20 (m, 8H), 6.20 (t, 1H, J=6.6 Hz), 5.30 (d, 1H, J=4.8 Hz), 4.28-4.21 (m, 1H), 3.90-3.86 (m, 1H), 3.22 (dd, 1H, J=5.4, 7.5 Hz), 3.13 (dd, 1H, J=3.0, 7.5 Hz), 2.45 (s, 6H), 2.28-2.19 (m, 1H), 2.16-2.08 (m, 1H), 1.54 (d, 3H, J=1.2 Hz), 0.21 (s, 9H)

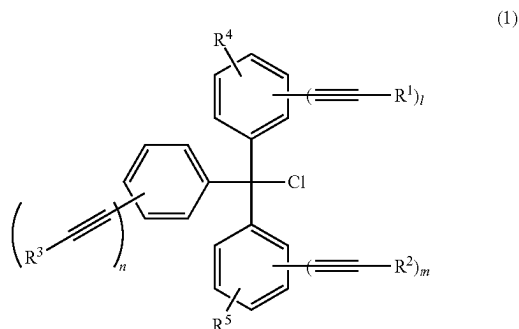
[0119] ¹³C-NMR (DMSO-d₆): δ 163.6, 150.4, 144.7, 139.3, 139.0, 137.5, 135.8, 131.3, 128.9, 128.8, 128.1, 125.3, 120.9, 109.6, 104.8, 94.5, 85.6, 85.2, 83.7, 70.3, 64.2, 39.4, 14.1, 11.8, -0.1

INDUSTRIAL APPLICABILITY

[0120] The trityl-based alcohol protecting group of the present invention comprises a DMTr or like protecting group and at least one triple bond introduced into the protecting group. The trityl-based alcohol protecting group of the present invention is a novel trityl-based alcohol protecting group that is useful in synthesizing nucleic acids. The trityl-based alcohol protecting group of the compound represented by formula (1) of the present invention is excellent in controlling the ease of deprotection.

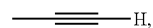
[0121] In the trityl-based alcohol protecting group of the compound represented by formula (1) of the present invention, a protecting group, such as a DMTr protecting group, changes in stability of the cation and forms a trityl-based alcohol protecting group that is easily or instantly removed, for example, under weak acidic conditions. The trityl-based alcohol protecting group of the compound represented by formula (1) of the present invention is excellent in that the triple bond introduced is detectable by Raman spectroscopy and various reactions can be tracked based on the trityl-based alcohol protecting group.

1. A compound represented by the following formula (1):

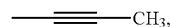


wherein

R¹, R², and R³ are same or different and each represent a hydrogen atom, an alkyl group, a trimethylsilyl group, an acetyl group, a benzoyl group, an ester group, an alkyl ester group, an aryl group, an alkyl alcohol group, a nitrile group, a group represented by



or a group represented by



l, m, and n each represent an integer of 0, 1, or 2, provided that l, m, and n are not 0 at the same time, and

R⁴ and R⁵ are the same or different and each represent an electron-donating group.

* * * * *